

1. What is the difference between API and MCP? Also, write the drawbacks of API over MCP.

API (Application Programming Interface) is a software-to-software bridge defining explicit rules, endpoints, methods, and parameters for software applications to exchange data and request service execution.

MCP (Model Context Protocol) is an open protocol (created by Anthropic) that standardizes how AI Large Language Models (LLMs) dynamically discover, fetch context, and execute tools across external services.

Practical Example Scenario

Goal: An AI Assistant needs to query your local database to list open bugs and summarize them.

- **The API Way:** You write custom backend/frontend code to handle authentication, execute specific database queries or API requests, parse the custom JSON payload, format the text, and inject it into the prompt. If the API schema changes, your code breaks and must be updated manually.
- **The MCP Way:** A single PostgreSQL **MCP Server** exposes your database tables as standard *resources* and query actions as *tools*. Any **MCP Client** (e.g., Cursor, Claude Desktop, or an AI Agent) automatically discovers the database capability at runtime and queries it directly using natural language—no custom glue code required.

Primary Drawbacks of APIs over MCP (for AI/LLM Applications)

1. **Exponential Integration Overhead ($N \times M$ Problem):** Connecting N AI tools/apps to M data sources require writing and maintaining $N \times M$ custom API wrappers. MCP reduces this to $N + M$ by providing a universal protocol.
2. **Lack of Dynamic Agent Discovery:** Standard APIs cannot natively announce their capabilities or tools in a structured format that LLMs understand out-of-the-box. Developers must manually construct tool/function calling definitions for every API endpoint.
3. **High Maintenance & Fragility:** Any breaking change in an API's schema or endpoint structure requires updating code across every client application. MCP abstracts the underlying implementation into standardized interfaces.
4. **Unstructured Context Delivery:** Standard APIs return raw data payloads that often exceed context window limits or require manual prompting logic to parse. MCP natively standardizes resource streaming, prompt templates, and context boundaries.